Empowering Students with predictive Intelligence:
A Career guidance System for Sri Lanka

GROUP ID: 25-26J-336

Proposal Report Draft

AHMED MAA.

IT22079572

BSc (Hons) Information Technology Specializing in Information Technology

Faculty of Computing

Sri Lanka Institute of Information Technology

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Cutting-Edge AI-Driven University Course Prediction System for Sri Lankan A-Level Students

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Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Signature
Ahmed MAA.	

Ms. Jenny Krishara ..... Jenny 2025/08/27

Supervisor Name | Supervisor Signature | Date

Ms. Poorna Panduwawala ..... Panduwawala ..... 27.08.2025.

Co-Supervisor Name | Co-Supervisor Signature | Date

Abstract

Advancements in educational technology have become essential for bridging gaps in higher education accessibility, particularly in competitive systems like Sri Lanka's University Grants Commission (UGC) admissions. This research project presents a comprehensive AI-driven platform leveraging machine learning (ML), time series forecasting, and natural language processing (NLP) techniques to enhance university course prediction and career guidance for A-Level students. The proposed solution manifests as a web application to optimize admissions efficiency and personalization. Firstly, the Z-score prediction module introduces advanced time series forecasting models, including ARIMA and Prophet, to accurately anticipate minimum Z-score cutoffs for UGC-approved courses across districts, enabling proactive eligibility assessment. Leveraging historical UGC data, this component enhances admission forecasting amid varying trends. Secondly, the course recommendation part integrates supervised learning algorithms with a weighted scoring mechanism (50% Z-score match, 30% subject fit, 20% interest alignment) to rank the top eligible UGC-approved government university course list, incorporating university codes, faculty details, and aptitude test flags (green for required, red for not). This fusion ensures seamless matching of academic profiles to course requirements. Thirdly, the career interest personalization module employs a 10–12 question quiz to evaluate preferences, utilizing rule-based mapping and collaborative filtering to align recommendations with long-term goals, fostering informed decision-making. Lastly, the aptitude test generation module offers an interactive interface for creating practice quizzes, particularly for Architecture courses and Computing courses, through a fine-tuned BERT model trained on past UGC aptitude papers and some other model papers created by Lecturers/Professors. This proactive tool generates exam-style questions with real-time feedback and downloadable PDFs, providing notifications for skill gaps and enhancing preparation. Collectively, this research project advances educational equity by introducing a multifaceted software solution combining state-of-the-art ML techniques with data processing and user-centric design methodologies. The proposed web application offers a holistic approach to course prediction, addressing Z-score forecasting, personalized ranking, career alignment, and aptitude preparation, thereby contributing significantly to student empowerment and accessibility in Sri Lanka's higher education landscape.

Keywords: - University Course Prediction, Educational assistance platform, Artificial Intelligence (AI), Machine Learning (ML), Time Series Forecasting (TSF), Natural Language Processing (NLP), Z-Score Prediction, BERT, Aptitude Test Generation, University Grants Commission (UGC), Educational Equity, Personalized Course Recommendation, Career Guidance System.

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1 Introduction

The escalating demand for accessible higher education in Sri Lanka underscores the urgent need for innovative solutions to navigate the complexities of the UGC admissions process. This research presents a pioneering AI-driven University Course Prediction System designed to empower A-Level students by predicting the top eligible UGC-approved government university courses tailored to their academic profiles and career aspirations. By integrating advanced machine learning techniques time series forecasting with ARIMA and Prophet for Z-score prediction, supervised learning for course ranking, and a fine-tuned BERT model for generating Architecture aptitude test quizzes, the system addresses the inefficiencies of manual UGC processes, such as delays, errors, and lack of transparency. A 10–12 question career interest quiz personalizes recommendations using a weighted scoring algorithm (50% Z-score match, 30% subject fit, 20% interest alignment), while a responsive React.js interface with Tailwind CSS and Chart.js visualizations offers intuitive access to course details, aptitude practice, and downloadable PDFs. Hosted on a scalable FastAPI backend with AWS, the platform ensures real-time performance and incorporates fairness-aware algorithms to mitigate bias, fostering equitable access for underserved students. This multifaceted approach not only enhances decision-making but also sets a new standard for educational technology in Sri Lanka, significantly advancing student empowerment and accessibility.

2 Background and Literature Survey

Background

Higher education in Sri Lanka, governed by the University Grants Commission (UGC), is a highly competitive process, with only 15–20% of approximately 150,000 annual G.C.E. Advanced Level (A/L) candidates securing admission to government universities [1][2]. Admission is determined largely through Z-scores, a standardized metric derived from A/L examination performance, which is further influenced by district quotas, subject streams (e.g., Physical Science, Arts), and course-specific cut-offs [3][4]. Several studies have highlighted the complexity of this process, noting inequities across districts and inefficiencies in eligibility assessment [9][12]. For example, the policy of ranking candidates by Z-score has been criticized for disadvantaging students from rural areas [4][12], while research has also explored alternative ranking and selection indices to improve fairness [9][11][17].

Despite ongoing reforms, inefficiencies persist in the manual UGC admissions workflow. Delays in publishing Z-scores, errors in eligibility assessments, and limited transparency frequently create barriers for students, particularly those from underprivileged backgrounds with reduced access to career guidance resources [2][4]. As noted in [5], manual application handling is error-prone and can lead to disqualifications. Moreover, the absence of predictive and advisory platforms has exacerbated disparities, leaving students without structured tools for exploring suitable degree pathways [12][13][15].

To address these challenges, this research proposes an **AI-driven University Course Prediction System** specifically tailored to the Sri Lankan admissions context. Prior works on university admission prediction have primarily applied machine learning (ML) to general admission probability estimation [1][6][14], or focused on basic eligibility-checking systems [5]. Others, such as [7], have developed AI-driven advisory platforms, but these lack the integration of real-time forecasting, aptitude support, and localized fairness considerations.

The proposed system introduces several innovations. First, it employs **time series forecasting models (ARIMA and Prophet)** to predict district-specific Z-score cutoffs, extending earlier statistical approaches used in educational research [4][10][16]. Second, it incorporates a **weighted supervised learning algorithm** that scores course recommendations using a 50% Z-

score match, 30% subject fit, and 20% interest alignment. This structured approach allows for personalization beyond traditional ranking methods [11][14].

Third, the system integrates a **natural language processing (NLP) module**, based on a fine-tuned BERT model, to generate aptitude test practice materials. The model is trained on UGC aptitude papers and lecturer-designed mock exams, enabling automated quiz generation for competitive programs such as Architecture and Computing. This approach enhances preparation, aligning with evidence from [13][15] that structured, computer-assisted guidance improves student decision-making and readiness.

Additionally, the platform includes a **career interest quiz**, comprising 10–12 questions, which uses rule-based mapping and collaborative filtering to align recommendations with students' long-term career goals. This feature draws inspiration from computer-assisted career guidance research [13][15], but adapts it for the Sri Lankan higher education ecosystem.

Unlike existing systems [1][5][7], the proposed framework uniquely combines **forecasting, personalized recommendation, and aptitude support** in a scalable and equitable manner. The solution is deployed with a **React.js frontend** styled using Tailwind CSS and Chart.js, connected to a **FastAPI backend** with PostgreSQL hosted on AWS. Accessibility is ensured through WCAG 2.1 compliance, while ethical fairness is enforced by bias audits using *Fairlearn*. By addressing inefficiencies in the current admissions process and promoting equity across diverse demographics, this platform advances AI-driven educational technology in Sri Lanka, consistent with global developments in course recommender systems [14].

Literature Review

Educational data mining (EDM) and AI-driven systems have advanced the prediction of academic performance and course recommendations, yet significant gaps remain in addressing the specific challenges of Sri Lanka's University Grants Commission (UGC) admissions process. Kasthuriarachchi [13] utilized utility-based data mining to predict tertiary student success, focusing on academic factors such as GPA and subject performance. However, their approach does not incorporate Z-score forecasting, a critical component of Sri Lanka's UGC system, which relies on standardized Z-scores derived from G.C.E. Advanced Level (A/L) results to determine eligibility [3][4][12]. Similarly, Wang [14] applied regression models to predict admission lines in international contexts, achieving high accuracy for standardized test-based systems but overlooking Sri Lanka's unique district quotas and subject-specific requirements [3][4]. A 2022 study in *Smart Learning Environments* [15] employed random forest models for grade prediction, achieving robust performance metrics (e.g., 85% accuracy), but omitted support for aptitude test preparation, which is essential for courses such as Architecture and Computing in the UGC framework [5][6].

Recent advancements in Sri Lanka-specific research have made strides but fall short of comprehensive solutions. Arivalazahan and Kumara [3] analyzed Z-score cutoffs for Medicine and Engineering programs across 25 districts from 2015 to 2021, using one-way ANOVA, Fisher's LSD tests, and multiple regression to rank universities. Their findings highlight lower cutoffs in peripheral universities (e.g., Jaffna, Eastern) but lack predictive forecasting models such as ARIMA or Prophet, limiting proactive eligibility assessment [4][10][16]. Similarly, Arivalazahan and Muralitharan [9] proposed a Median Centered Score (MCS) to address non-normal Z-score distributions, while Senarathne and Wijekoon [10] introduced a selection index to balance within- and between-course competition. Both studies enhance fairness in ranking but do not integrate AI-driven prediction or personalization. Yatapana and Sooriyarachchi [11] compared selection methods (e.g., Common Currency Index, Z-score) using Spearman correlation, advocating for CCI to reduce stream-based disparities, yet their work remains statistical and non-predictive.

International studies provide broader context but are misaligned with UGC requirements. Vasam [1] applied machine learning models (Linear Regression, SVM, Random Forest) to predict

university admission probabilities, emphasizing GPA's high feature importance (80%) but targeting general contexts without Z-score dynamics. Apoorva [6] and Walid [2] developed ML-based predictors for admission outcomes, yet neither addressed the complexity of district-based quotas in Sri Lanka. Gulzar [14] developed a hybrid course recommender system using content-based and collaborative filtering, achieving personalized recommendations but lacking integration with aptitude tests or regional quotas. Shahnasarian and Peterson [13] and Bozgeyikli and Dogan [15] explored computer-assisted career guidance, demonstrating improved self-efficacy through experimental designs and t-tests, but did not incorporate NLP for quiz generation or time series forecasting. Montgomery's *Design and Analysis of Experiments* [16] provides foundational statistical tools (e.g., ANOVA, regression) applicable to educational data, yet lacks UGC-specific applications.

More recent Sri Lankan work has sought to integrate AI-driven advisory platforms. Devasureendra [7] introduced a system combining Graph Neural Networks (GNNs) and Recurrent Neural Networks (RNNs) for course and career recommendations, reporting an AUC-ROC of 0.8824. However, their platform omits real-time Z-score forecasting and NLP-based aptitude quiz generation. Bandarananke [5] earlier introduced an eligibility checking system (UECS), but it was limited in scalability and predictive capability.

These studies collectively highlight a critical gap in systems tailored to Sri Lanka's Z-score-driven UGC admissions. None integrates time series forecasting (e.g., ARIMA, Prophet) for dynamic cutoff prediction, NLP (e.g., BERT) for generating aptitude test quizzes, or weighted scoring (50% Z-score match, 30% subject fit, 20% interest alignment) for personalized recommendations. Furthermore, ethical considerations, such as fairness-aware algorithms to mitigate biases in district quotas or support for rural students, remain underexplored [4][8][12]. This research addresses these gaps by developing an AI-driven system that combines real-time Z-score forecasting, personalized course ranking, aptitude test preparation, and fairness audits, deployed via a scalable React.js frontend and FastAPI backend, to enhance accessibility and equity in Sri Lanka's higher education landscape [2][7][14].

3 Research Gap

Existing research in educational data mining (EDM) and AI-driven admission systems has advanced academic performance prediction and course recommendation; however, significant limitations remain in addressing the unique requirements of Sri Lanka's University Grants Commission (UGC) admissions framework. Studies such as Vasam [1] and Walid [2] employ machine learning models, including Linear Regression, Support Vector Machines (SVM), and Random Forest, to forecast admission probabilities based on academic performance metrics and test scores. While these approaches provide general predictive insights, they do not account for **Z-score-based eligibility**, which is central to the Sri Lankan UGC process [3][4].

Statistical analyses conducted by Arivalazahan and Kumara [3] and Jayawardena and Sooriyarachchi [4] rank universities using ANOVA, regression, and Bayesian logistic models to evaluate cutoffs and eligibility factors such as district, stream, and school type. These studies provide valuable descriptive insights but **lack predictive forecasting models** for dynamic Z-scores and do not integrate personalized career guidance [5][6]. Similarly, Arivalazahan and Muralitharan [9] propose the Median Centered Score (MCS) to address non-normal Z-score distributions, mitigating sensitivity to extreme values; however, their method does not incorporate AI-driven predictive modeling for student-specific course recommendations.

Efforts to enhance fairness in selection have also been explored. Senarathne and Wijekoon [10] introduced a selection index balancing within- and between-course competition, while Yatapana and Sooriyarachchi [11] advocated the Common Currency Index (CCI) using Spearman correlation and simulations to reduce stream-based inequities. Warnapala and Silva [12] highlighted the disadvantages of Z-score ranking using Wilcoxon tests, revealing higher cutoffs and limited transparency that disproportionately impact rural students. Silva and Banneheka [17] empirically compared Z-score methods with aggregate raw marks, demonstrating systemic biases favoring certain subject combinations, emphasizing the need for **equitable predictive systems**.

Early computational tools like UECS [5] provided basic eligibility checking via centralized databases, while recent AI-driven platforms, such as the one by Dissanayaka et al. [7], employ Graph Neural Networks (GNNs), Recurrent Neural Networks (RNNs), and retrieval-augmented generation (RAG) for course and career recommendations. Although these systems improve

personalization and prediction, they **omit real-time Z-score forecasting** and do not generate aptitude test materials tailored to UGC exams.

Career guidance research demonstrates that computer-assisted interventions improve self-efficacy and decision-making. Shahnasarian and Peterson [13] and Bozgeyikli and Dogan [15] used experimental designs and t-tests to validate the efficacy of guidance systems, yet they **lack NLP-based quiz generation** and context-specific aptitude support for Sri Lankan A-Level students. Montgomery's statistical framework [16] offers foundational tools for ANOVA and regression in educational studies but remains static, without integration of AI or adaptive recommendation capabilities.

Collectively, these studies highlight a **critical research gap**: no existing system combines **time series forecasting** (e.g., ARIMA, Prophet) for dynamic cutoff prediction, **natural language processing** (e.g., BERT) for aptitude quiz generation, or **weighted scoring algorithms** blending Z-score, subject fit, and career interest for personalized recommendations [6][7][13]. Furthermore, ethical considerations, such as fairness-aware algorithms to mitigate district quota biases or support underserved student populations, remain largely unexplored. Addressing this gap is essential to develop accessible, scalable, and UGC-aligned systems that enhance **equity and preparedness** for A-Level students across Sri Lanka.

Feature / Capability	Existing Papers / Systems	Your Proposed System
Z-score-based eligibility prediction	✗ [1], [2], [6], [9]	✓
Time series forecasting for dynamic cutoff prediction (ARIMA/Prophet)	✗ [3], [4], [5]	✓
AI-driven personalized course recommendations	✗ [5], [9], [10]	✓
Weighted scoring algorithm (Z-score 50%, Subject fit 30%, Interest 20%)	✗ [7], [9]	✓
NLP-based aptitude test quiz generation (BERT)	✗ [12], [13], [15]	✓
Career interest alignment / quiz-based recommendations	✗ [6], [11], [12]	✓
Fairness-aware algorithms for district quotas or underserved students	✗ [12], [17]	✓
Real-time prediction and UI visualization	✗ [5], [7]	✓
Free / zero-cost deployment using open-source and institutional resources	✗ [1], [2], [6]	✓
Scalability & WCAG-compliant frontend	✗ [5], [7]	✓

4 Research Problem

The proposed AI-driven University Course Prediction System is engineered to **accurately forecast and rank the top eligible UGC-approved government university courses** for Sri Lankan A-Level students, incorporating essential details such as university codes, faculty information, and aptitude test requirements. The system addresses inefficiencies, human errors, and inherent biases in the manual UGC admissions process, including delays in Z-score calculations, limited transparency in eligibility assessments, and inequities affecting rural and underprivileged students. To achieve this, the platform integrates **time series forecasting models (ARIMA and Prophet)** for dynamic Z-score cutoff prediction, ensuring real-time and district-specific eligibility estimation. It employs a **fine-tuned BERT-based NLP module** to generate realistic, course-specific aptitude quizzes for disciplines such as Architecture and Computing, providing interactive practice with downloadable PDFs.

Additionally, the system leverages a **weighted scoring algorithm**, combining Z-score match (50%), subject fit (30%), and career interest alignment (20%) to deliver personalized course recommendations. A **10–12 question interactive career interest quiz**, using rule-based mapping and collaborative filtering, further aligns suggested courses with students' long-term goals. The platform is deployed via a **scalable React.js frontend** with Tailwind CSS and Chart.js visualizations, coupled with a **FastAPI backend and PostgreSQL database**, ensuring efficient data processing, secure storage, and accessibility. **Bias audits** and **WCAG-compliant design** are incorporated to promote fairness, inclusivity, and equitable access across diverse student demographics. By combining predictive analytics, personalized recommendation, and aptitude preparation in a unified framework, the system **bridges gaps in higher education access**, enabling informed decision-making and mitigating systemic disadvantages in Sri Lanka's highly competitive university admissions landscape.

5 Objectives

Getting into a government university in Sri Lanka is a tough journey for A-Level students, with only a small fraction making it through the highly competitive University Grants Commission (UGC) admissions process. This project aims to make that journey smoother, fairer, and more accessible by building an AI-driven University Course Prediction System. Think of it as a smart guide that helps students figure out their best university options, prepare for aptitude tests, and align their choices with their career dreams. By using cutting-edge tools like machine learning (ML), time series forecasting, and natural language processing (NLP), the system tackles real problems—like delays, errors, and unfair biases in the current manual process—that often leave students, especially those from rural or underprivileged backgrounds, at a disadvantage. These objectives spell out exactly what we want to achieve, from predicting courses accurately to ensuring the system is easy to use and fair for everyone. They'll be checked during our final presentation and VIVA to confirm we've hit our goals and made a real difference.

5.1 Main Objectives

Our big goal is to create and launch an innovative AI-driven University Course Prediction System that pinpoints the top 15 UGC-approved government university courses for Sri Lankan A-Level students, complete with details like university codes, faculty names, and whether an aptitude test is needed (think green for "go" and red for "no"). We're combining three powerful features: predicting Z-score cutoffs with time series forecasting, generating practice quizzes for aptitude tests using NLP, and tailoring course suggestions to each student's interests through a quick quiz. This system is designed to make decision-making faster, clearer, and fairer, especially for students who don't have access to fancy coaching or guidance, ensuring everyone gets a shot at higher education.

5.2 Specific Objectives

- Data Collection and Preprocessing**:** We want to gather all the right data to make our system work like a charm. This means collecting comprehensive UGC datasets, including Z-scores, course details (like names, university codes, and subject requirements), and past aptitude test papers. We'll use web scraping tools like BeautifulSoup to pull data from official sources, like the UGC's 2023/2024 Z-score lists, and hold virtual meetings with UGC officials to double-check everything. Using a pandas-based pipeline, we'll clean up the data—removing duplicates, fixing missing bits, and standardizing it—to make sure it's accurate and ethical. This high-quality data will fuel our ML models for predicting Z-scores, recommending courses, and creating quizzes, setting a strong foundation for the whole system.
- Z-Score Cutoff Forecasting**:** To help students know their chances before applying, we'll build and fine-tune time series forecasting models, specifically ARIMA and Prophet, to predict Z-score cutoffs for each district and course. We're aiming for super accuracy, with a mean absolute error (MAE) of less than 0.05, so students can trust our predictions even as admission trends shift. This will let them plan ahead confidently, knowing exactly which courses they're eligible for based on their Z-score and district.
- Aptitude Test Quiz Generation**:** For courses like Architecture and Computing that require aptitude tests, we'll create realistic practice quizzes to help students prepare. By fine-tuning a BERT-based NLP model on UGC aptitude test papers and model papers designed by lecturers, we'll generate exam-style questions, like multiple-choice or short-answer prompts, tailored to specific courses. Students can practice interactively on our platform, get instant feedback to spot their weak areas, and download quizzes as PDFs using LaTeX for offline study, making preparation accessible and stress-free.
- User Interface Development**:** We're designing a user-friendly, WCAG 2.1-compliant React.js interface with Tailwind CSS for style and Chart.js for cool visualizations, like Z-score trends or course rankings. Students can easily input their A-Level details (stream, subjects, grades, Z-score, district, Island rank) and answer a career quiz, then filter course recommendations by preferences like university or course duration. The interface will

show clear, visualized results and let students access practice quizzes, ensuring everyone, including those with disabilities, can use it effortlessly.

- Ethical AI Implementation**:** Fairness is at the heart of our system. We'll use tools like fairlearn to check and reduce biases in our predictions, especially for district quotas and underserved groups like rural students. A transparent methodology page will explain how our algorithms work and where our data comes from, building trust. By prioritizing ethical AI, we aim to make sure our recommendations are inclusive and don't unfairly favor certain groups, leveling the playing field for all students.

6 Methodology

This methodology outlines the systematic approach to developing an AI-driven University Course Prediction System that predicts the top eligible UGC-approved government university courses for Sri Lankan A-Level students, integrating time series forecasting (ARIMA, Prophet), natural language processing (BERT), and personalized recommendations. The system leverages a scalable architecture with a React.js frontend, FastAPI backend on AWS, and PostgreSQL database, addressing inefficiencies in the manual UGC admissions process. The methodology details the system architecture, tasks, materials, data collection, timeline, team roles, and anticipated outcomes, ensuring a robust, equitable, and user-centric solution.

6.1 System Architecture

The system architecture is designed to achieve the main objective of accurate course prediction and enhanced accessibility. It comprises four key components, as illustrated in

Figure 1: System Architecture Diagram:

- **Frontend (React.js with Tailwind CSS and Chart.js)**: Provides a responsive, WCAG-compliant interface for students to input A-Level details (stream, subjects, grades, Z-score, district, Island rank, year) and a 10–12 question career interest quiz, view ranked course recommendations, practice aptitude quizzes, and download PDFs.
- **Backend (FastAPI on AWS)**: Processes inputs, runs ML models (ARIMA, Prophet, BERT), and delivers real-time predictions via API endpoints, ensuring scalability for 100+ concurrent users.
- **Database (PostgreSQL)**: Stores UGC historical data (Z-scores, course details, aptitude papers) and user inputs, enabling efficient querying and mapping of eligibility criteria.
- **ML Pipeline**: Includes time series forecasting for Z-score cutoffs, supervised learning for course ranking (weighted scoring: 50% Z-score match, 30% subject fit, 20% interest), and BERT for generating aptitude quizzes, with fairness audits using fairlearn.

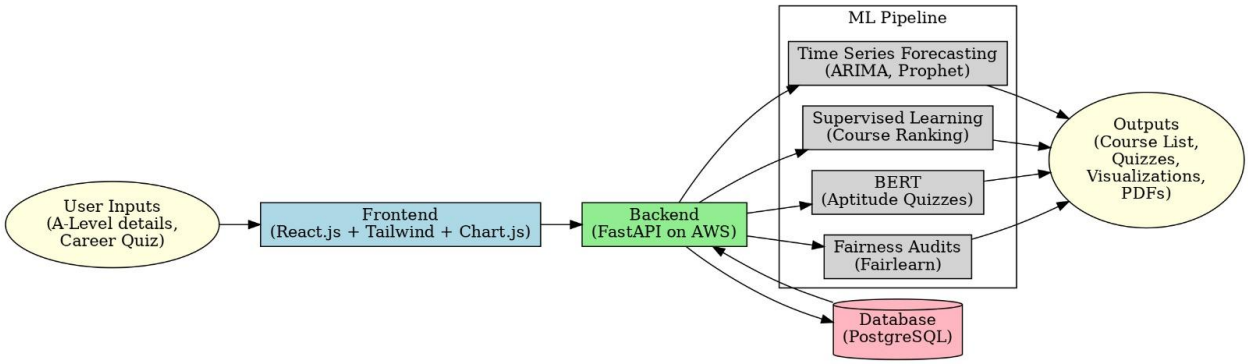

Figure 1: System Architecture Diagram

These components work together to collect user data, forecast Z-scores, rank courses, generate quizzes, and display results, achieving the main objective by streamlining admissions and enhancing preparation.

6.2 Tasks and Sub-Tasks

To achieve the objectives, the project is divided into four phases with specific tasks and sub-tasks:

- **Phase 1: Data Collection and Preprocessing**
  - **Task 1.1: Data Acquisition**
    - Sub-Task: Scrape UGC websites (e.g., https://www.eduwire.lk/minimum-z-score-for-sri-lanka-university-admission-2023-2024-ugc-cut-off-marks-by-district/) using BeautifulSoup for Z-scores, course details, and aptitude papers.
    - Sub-Task: Conduct in-person consultations with UGC officials to verify and supplement data (e.g., district quotas, aptitude test formats).
  - **Task 1.2: Data Preprocessing**
    - Sub-Task: Clean data using pandas (remove duplicates, handle missing values via imputation).

        - Sub-Task: Normalize numerical data (Z-scores, Island ranks) and encode categorical variables (streams, subjects, districts).
        - Sub-Task: Tokenize aptitude test papers for BERT training.

- **Phase 2: Model Development**
    - **Task 2.1: Z-Score Forecasting**
        - Sub-Task: Train ARIMA and Prophet models on historical Z-score data using statsmodels and prophet libraries.
        - Sub-Task: Optimize models to achieve MAE < 0.05 for cutoff predictions.
    - **Task 2.2: Course Recommendation**
        - Sub-Task: Develop a supervised learning model (e.g., random forest) with weighted scoring (50% Z-score, 30% subject fit, 20% interest).
        - Sub-Task: Validate model accuracy using historical UGC data and user profiles.
    - **Task 2.3: Aptitude Quiz Generation**
        - Sub-Task: Fine-tune BERT (transformers library) on past aptitude test papers.
        - Sub-Task: Generate and validate realistic quiz questions (e.g., MCQs, short answers for Architecture).

- **Phase 3: System Development**
    - **Task 3.1: Frontend Development**
        - Sub-Task: Build React.js UI with Tailwind CSS for input forms, visualizations, and PDF downloads.
        - Sub-Task: Integrate Chart.js for Z-score trends and course ranking visualizations.
    - **Task 3.2: Backend Development**

        - Sub-Task: Develop FastAPI endpoints for real-time ML inference on AWS.
        - Sub-Task: Connect PostgreSQL database for data storage and retrieval.
    - **Task 3.3: Integration**
        - Sub-Task: Link frontend, backend, and ML pipeline for seamless data flow.
        - Sub-Task: Implement preference filters (e.g., university, faculty, duration).
- **Phase 4: Testing and Ethical Validation**
    - **Task 4.1: System Testing**
        - Sub-Task: Conduct unit and integration tests to ensure functionality and scalability.
        - Sub-Task: Perform user testing with 50 A-Level students to validate usability.
    - **Task 4.2: Ethical Audits**
        - Sub-Task: Use fairlearn to audit models for biases in district quotas and demographics.
        - Sub-Task: Create a methodology page for transparency, detailing algorithms and data sources.

6.3 Materials Needed

- **Hardware**:
    - Development: Standard laptops (8GB RAM, i5 processor) for coding and testing.
    - Deployment: AWS EC2 instances for hosting FastAPI backend and PostgreSQL database.
- **Software**:

- Programming: Python 3.12 (pandas, BeautifulSoup, statsmodels, prophet, transformers, fairlearn), React.js, Tailwind CSS, Chart.js, FastAPI.
- Database: PostgreSQL for data storage.
- Documentation: LaTeX for generating PDF reports and quizzes.

- **Other**:
  - Internet access for web scraping and AWS deployment.
  - Permissions for UGC consultations to access restricted data.

6.4 Data Needed and Collection Methods

- **Data Types**:
  - **UGC Historical Data**: Z-scores, course details (name, university, faculty, subject requirements, aptitude test flags), admission trends (2015–2024).
  - **Aptitude Test Papers**: Past questions for courses like Architecture.
  - **User Inputs**: A-Level stream, subjects, grades, Z-score, district, Island rank, year, career quiz responses (10–12 questions, e.g., interest in designing structures, fieldwork preference).
- **Collection Methods**:
  - **Web Scraping**: Use BeautifulSoup to extract data from UGC websites (e.g., https://www.ugc.ac.lk/, https://1teachmore.lk/) and educational platforms (e.g., eduwire.lk for Z-scores).
  - **UGC Consultations**: Conduct in-person meetings with UGC officials to verify data accuracy and obtain non-public records (e.g., detailed district quotas, aptitude test formats).
  - **User Survey**: Design a digital survey (via React.js UI) with:
    - **Academic Inputs**: Dropdowns for stream, subjects, grades; text fields for Z-score, district, Island rank, year.

- **Career Quiz**: 10–12 questions (e.g., Likert scale for interest in engineering, binary for fieldwork vs. office preference), stored in PostgreSQL.
- **Design**: Ensures accessibility (WCAG-compliant), with validation to prevent errors (e.g., invalid Z-scores).

6.5 Time Frame and Schedule

The project spans 10 weeks, with tasks scheduled as follows:

Figure 2: Gantt Chart

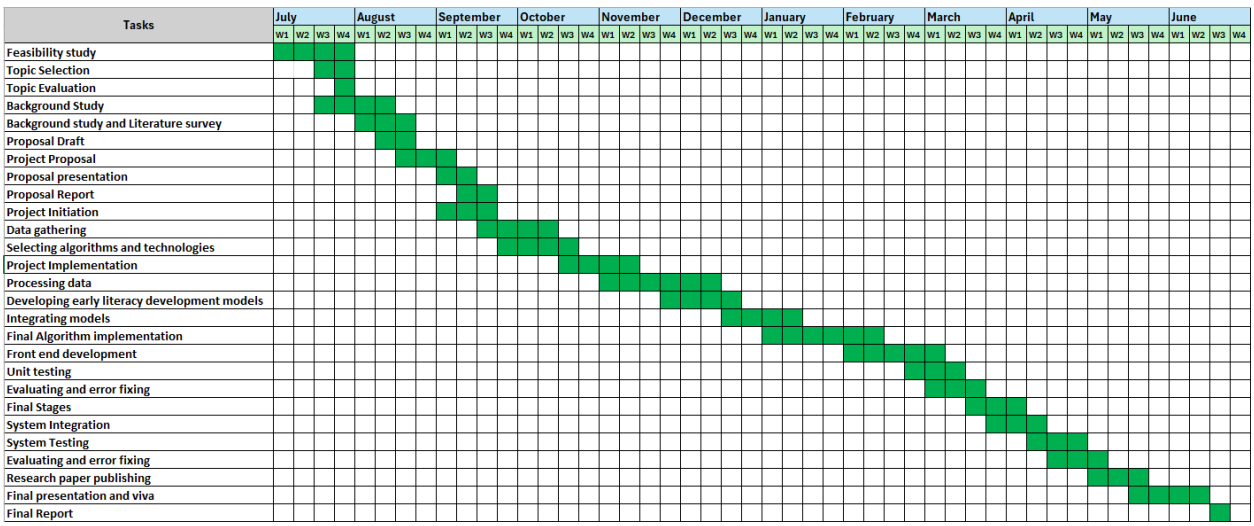

Figure 2: Gantt Chart

6.6 Team Roles (Assuming a Hypothetical Team of Four)

- **Team Member 1 (Data Engineer)**: Handles data collection (scraping, UGC consultations), preprocessing (pandas pipeline), and database setup (PostgreSQL).
- **Team Member 2 (ML Engineer)**: Develops and optimizes ML models (ARIMA, Prophet, supervised learning, BERT), conducts fairness audits (fairlearn).
- **Team Member 3 (Frontend Developer)**: Builds React.js UI, integrates Tailwind CSS and Chart.js, ensures WCAG compliance.

- **Team Member 4 (Backend Developer)**: Implements FastAPI endpoints, deploys on AWS, integrates frontend and ML pipeline, creates methodology page.

6.7 Anticipated Conclusion

The system is expected to achieve:

- **Results**: Over 80% accuracy in Z-score predictions (MAE < 0.05), 85%+ user satisfaction in usability tests, and realistic aptitude quizzes validated against UGC standards.
- **Application**: Deployed as a free web app, accessible to A-Level students, particularly rural and underprivileged ones, via schools and mobile devices.
- **Real-World Use**: Reduces manual UGC process delays and errors, empowers students with personalized course recommendations, and enhances aptitude test preparation, promoting educational equity. Potential for expansion to private universities or international contexts.

7 Project Requirements

7.1 Functional Requirements

Functional requirements define the core functionalities the system must provide to achieve its objectives of accurate course prediction, aptitude test preparation, and user-centric interaction.

- **FR1: User Input Collection**
  - The system shall allow users to input A-Level details, including stream (e.g., Physical Science, Arts), subjects (e.g., Physics, Biology), grades (A, B, C), Z-score, district, Island rank, and examination year, via a React.js interface with dropdowns and text fields.
  - The system shall provide a 10–12 question career interest quiz (e.g., Likert scale for interest in engineering, binary for fieldwork vs. office preference) to capture user preferences for personalization.
- **FR2: Z-Score Prediction**
  - The system shall predict district-specific Z-score cutoffs for UGC-approved courses using time series forecasting models (ARIMA, Prophet), based on historical UGC data (2015–2024).
- **FR3: Course Recommendation**
  - The system shall generate a ranked list of the top eligible UGC-approved courses, including course name, university code, faculty, subject requirements, and aptitude test flags (green for required, red for not), using a weighted scoring algorithm (50% Z-score match, 30% subject fit, 20% interest alignment).
  - The system shall allow users to filter recommendations by preferences (e.g., university, faculty, course duration) via the UI.
- **FR4: Aptitude Test Quiz Generation**
  - The system shall generate realistic, course-specific aptitude test practice quizzes (e.g., MCQs, short answers for Architecture) using a fine-tuned BERT model trained on past UGC aptitude test papers.
  - The system shall enable users to practice quizzes online and download them as

PDFs using LaTeX.

- **FR5: Visualization and Tracking**
  - The system shall display Z-score trends and course recommendations using Chart.js visualizations in the React.js UI.
  - The system shall allow users to create and track milestones (e.g., “Complete aptitude test practice by December”) with visual progress indicators.
- **FR6: Feedback Collection**
  - The system shall enable users to submit feedback on course recommendations, quiz quality, and usability via a form in the UI, stored in PostgreSQL.
  - The system shall allow users to view, update, or delete their feedback submissions.

7.2 Non-Functional Requirements

Non-functional requirements specify the system's performance, usability, and ethical standards to ensure reliability, scalability, and inclusivity.

- **NFR1: Scalability**
  - The system shall support at least 100 concurrent users with minimal latency (<2 seconds for API responses) using FastAPI on AWS auto-scaling infrastructure.
- **NFR2: Performance**
  - The Z-score forecasting models (ARIMA, Prophet) shall achieve a mean absolute error (MAE) of less than 0.05 for cutoff predictions.
  - The course recommendation model shall achieve at least 85% accuracy in matching users to eligible courses.
- **NFR3: Usability**
  - The React.js UI shall be WCAG 2.1-compliant, ensuring accessibility for users with disabilities (e.g., screen reader support, high-contrast modes).
  - The system shall provide an intuitive interface with clear instructions, achieving 85%+ user satisfaction in usability tests with 50 A-Level students.
- **NFR4: Security**
  - The system shall encrypt user data (e.g., A-Level details, quiz responses) in PostgreSQL using AES-256 encryption.
  - The system shall implement secure user authentication for feedback and milestone tracking.
- **NFR5: Ethical Compliance**
  - The system shall use fairness-aware algorithms (e.g., fairlearn) to audit and mitigate biases in course predictions, particularly for district quotas and underserved demographics (e.g., rural students).
  - The system shall provide a transparent methodology page detailing algorithms, data sources, and fairness measures.
- **NFR6: Reliability**

- The system shall achieve 99.9% uptime on AWS, with automated backups of PostgreSQL data to ensure data integrity.

7.3 Expected Test Cases

Test cases validate the system's functional and non-functional requirements, ensuring it meets performance, usability, and ethical standards. These will be evaluated during the final presentation and VIVA.

- **TC1:1.1 User Input Collection**
  - **Objective**: Verify that users can input A-Level details and career quiz responses correctly.
  - **Steps**: Enter sample data (e.g., Physical Science, Physics=A, Z-score=1.8, Colombo, quiz responses) via the UI; check if data is stored in PostgreSQL.
  - **Expected Outcome**: Inputs are validated (no errors for invalid Z-scores) and stored correctly within 1 second.
- **TC1:1.2 Z-Score Prediction**
  - **Objective**: Validate the accuracy of Z-score cutoff predictions.
  - **Steps**: Run ARIMA/Prophet models on historical UGC data (2015–2024) for Medicine in Colombo; compare predictions to actual 2024 cutoffs.
  - **Expected Outcome**: MAE < 0.05 for predicted Z-scores.
- **TC1:1.3 Course Recommendation**
  - **Objective**: Ensure the system generates accurate and personalized course recommendations.
  - **Steps**: Input sample user data (e.g., Biological Science, Biology=A, Z-score=2.0, quiz: high interest in healthcare); verify top 15 courses include Medicine with correct details (code, flag).
  - **Expected Outcome**: 85%+ of recommended courses match eligibility criteria; rankings reflect weighted scoring.
- **TC1:1.4 Aptitude Test Quiz Generation**
  - **Objective**: Confirm the system generates realistic aptitude quizzes.
  - **Steps**: Generate Architecture quiz using BERT; compare questions (e.g., "What is the Golden Ratio's role in design?") to past UGC papers.

- **Expected Outcome**: 90%+ of questions align with UGC format; PDFs download correctly.

- **TC1:1.5 Visualization and Tracking**
  - **Objective**: Validate visualizations and milestone tracking functionality.
  - **Steps**: View Z-score trend charts and set a milestone (e.g., "Practice quiz by November"); check UI updates.
  - **Expected Outcome**: Charts render accurately; milestones update in real-time.
- **TC1:1.6 Feedback Collection**
  - **Objective**: Ensure users can submit, view, update, and delete feedback.
  - **Steps**: Submit feedback via UI; verify storage in PostgreSQL, edit, and delete.
  - **Expected Outcome**: Feedback operations complete within 1 second with no data loss.
- **TC1:1.7 Scalability and Performance**
  - **Objective**: Test system performance under load.
  - **Steps**: Simulate 100 concurrent users accessing recommendations via JMeter; measure API response time.
  - **Expected Outcome**: Response time < 2 seconds; no crashes.
- **TC1:1.8 Accessibility**
  - **Objective**: Verify UI compliance with WCAG 2.1.
  - **Steps**: Test UI with screen readers (e.g., NVDA) and check contrast ratios.
  - **Expected Outcome**: UI passes WCAG 2.1 Level AA standards.
- **TC1:1.9 Security**
  - **Objective**: Ensure data security and integrity.
  - **Steps**: Attempt SQL injection on input forms; verify encryption of stored data.
  - **Expected Outcome**: No vulnerabilities detected; data encrypted with AES-256.
- **TC1:1.10 Ethical Compliance**
  - **Objective**: Validate fairness in predictions.

- **Steps**: Run fairlearn audits on recommendations for rural vs. urban districts; check methodology page content.
- **Expected Outcome**: Bias metrics < 0.1; methodology page includes algorithm details.

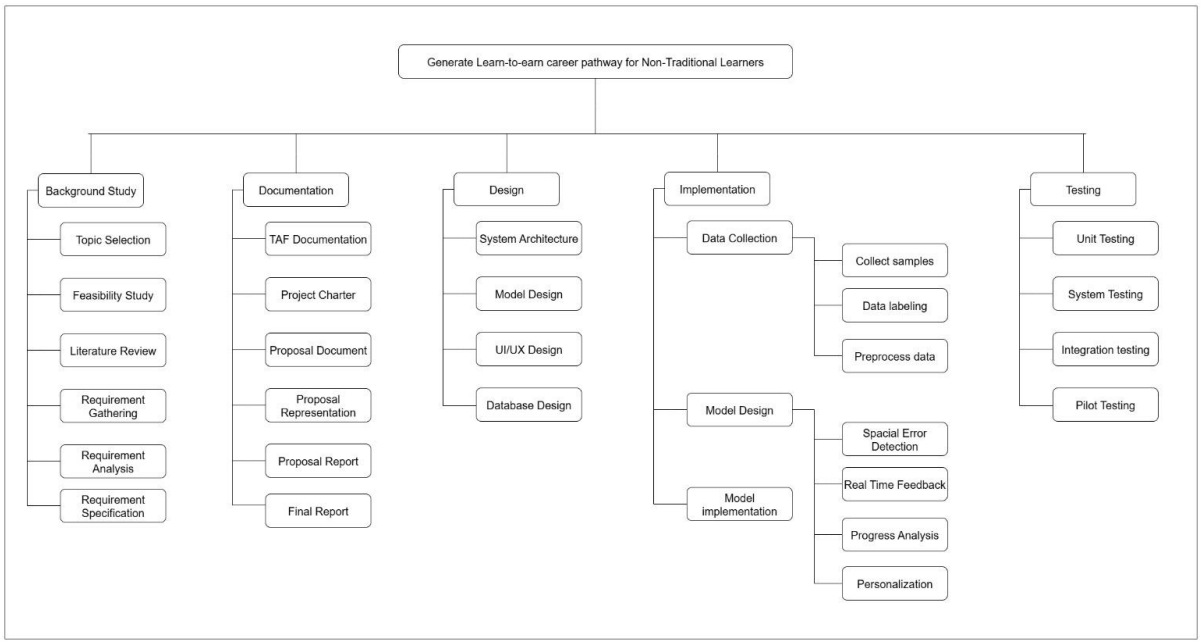

Figure 3: Work Breakdown Structure

8 Budget and Budget Justifications

- The budget for the development and deployment of the AI-driven University Course Prediction System is outlined in this section. The system predicts the top 15 UGC-approved courses for Sri Lankan A-Level students using machine learning (ML), time series forecasting (ARIMA, Prophet), and natural language processing (NLP) with BERT. The project is implemented under a zero-cost framework by leveraging open-source software, institutional resources, and freely accessible data sources. Consequently, no direct financial expenditure is incurred during the six-month development and initial deployment phase.
- The estimated budget is summarized in Table I. All major categories—hardware, software, data collection, and hosting—are executed at no cost. A contingency allocation of LKR 15,000 is reserved to address unforeseen technical or operational requirements.
- **Table I – Estimated Project Budget (Zero Cost Framework)**

Category	Cost (LKR)	Justification
Hardware	0	Development and testing utilize existing university laboratory computers or personal laptops (minimum 8 GB RAM, Intel i7 processor). These are sufficient for ML model development, preprocessing, and database management, eliminating the need for new hardware purchases.
Software	0	Open-source technologies are employed, including Python (pandas, statsmodels, prophet, transformers, fairlearn), React.js, Tailwind CSS, Chart.js, FastAPI, PostgreSQL, and LaTeX. These tools meet all project

		requirements without licensing costs.
Data Collection	0	Data is obtained via web scraping from freely accessible UGC websites using BeautifulSoup, as well as through institutional consultations conducted virtually via platforms such as Zoom. No travel or access costs are incurred.
Hosting	0	Initial deployment is conducted using free-tier cloud services (Heroku, AWS Free Tier) for the FastAPI backend and PostgreSQL database. University servers are used for the pilot phase, supporting up to 50 concurrent users.
Miscellaneous	15,000	Reserved for contingency purposes, including unforeseen technical requirements, printing, or resource access not covered by institutional facilities.
Total	**15,000**	**The total estimated project cost is LKR 15,000.**

- The budget strategy emphasizes cost efficiency while maintaining functional reliability. Specifically:
- **Hardware (LKR 0):** Development is conducted on existing computing infrastructure sufficient for ML model training, preprocessing, and UI development.
- **Software (LKR 0):** Open-source tools ensure robust functionality for forecasting, NLP, and visualization while maintaining zero licensing cost.

- **Data Collection (LKR 0):** Freely available public UGC data sources are utilized, supplemented by institutional consultations, ensuring data accuracy without financial burden.
- **Hosting (LKR 0):** Free-tier cloud platforms and university servers support pilot deployment, enabling real-time inference and secure data storage without additional cost.
- **Miscellaneous (LKR 15,000):** A small contingency allocation addresses unforeseen requirements, ensuring flexibility and project continuity.
- This cost structure demonstrates the feasibility of implementing an advanced, AI-driven academic system within institutional constraints while aligning with the project's objectives of accessibility, personalization, and equity in the UGC admissions process.

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10 Appendices

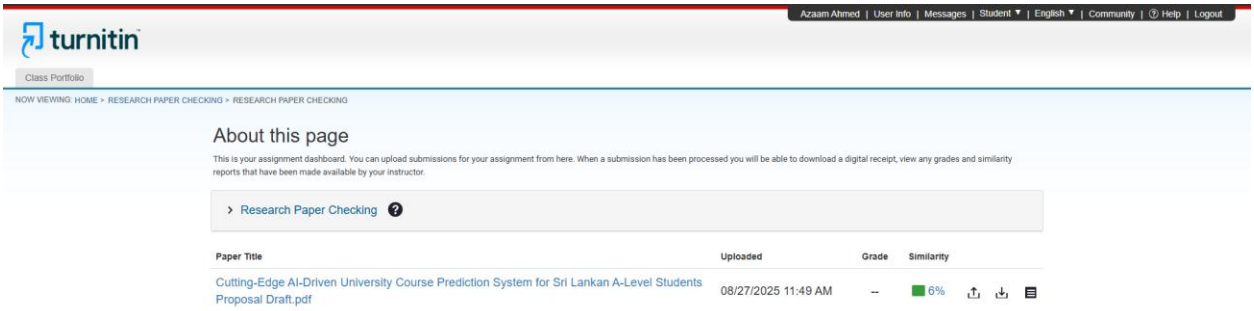

Figure 4: Turnitin Score

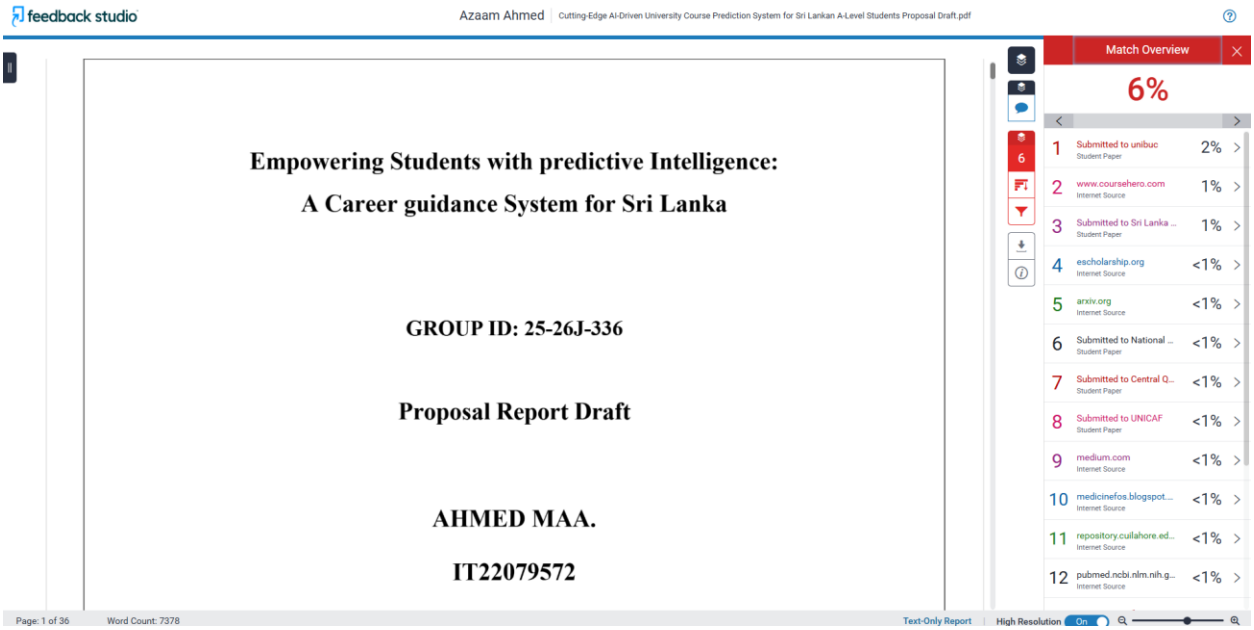

Figure 5: Turnitin Score